ELECTRICAL PERMIT COMPLIANCE DRAWING - PHILIPPINE ELECTRICAL CODE (PEC 2017 PART 1)

SERVICE: 230V AC, 1-PHASE, 2-WIRE + SOLID GROUND, 60 HZ | 10 kAIC MINIMUM SYMMETRICAL RMS

1. SCHEDULE OF LOADS & CIRCUIT COMPUTATION (LP-1 (MAIN DISTRIBUTION PANEL) - MAIN DISTRIBUTION PANEL)

BRANCH VOLTAGE DROP LIMIT: <= 3.0% (PEC SEC. 2.10.2.2(A) FPN 4) | FEEDER LIMIT: <= 2.0% (PEC SEC. 2.15.1.2)

CKT NO.	LOAD DESCRIPTION	VOLT	LOAD (VA)	CURRENT (A)	P	BREAKER	3D RUN	%VD	CONDUCTORS (THHN/THWN-2)	RACEWAY / CONDUIT	REMARKS / PEC COMPLIANCE
Ckt 1	Lighting Outlets (Living, Dining, Kitchen)	230V	1200	5.2	2P	15 AT	14.5m	0.82%	2 - 2.0 mm2 THHN Cu + 1 - 2.0 mm2 Ground	20 mm DIA PVC	PASS (<3%)
Ckt 2	Convenience Outlets (Living & Master BR)	230V	1800	7.8	2P	20 AT	22.0m	1.45%	2 - 3.5 mm2 THHN Cu + 1 - 3.5 mm2 Ground	20 mm DIA PVC	PASS (<3%)
Ckt 3	Air Conditioning Unit (Master BR - 1.5 HP)	230V	2400	10.4	2P	30 AT	38.5m	2.12%	2 - 5.5 mm2 THHN Cu + 1 - 5.5 mm2 Ground	20 mm DIA PVC	PASS (<3%)
Ckt 4	Induction Cooktop / Range (Kitchen)	230V	4500	19.6	2P	40 AT	18.0m	1.68%	2 - 8.0 mm2 THHN Cu + 1 - 5.5 mm2 Ground	25 mm DIA PVC	PASS (<3%)
Ckt 5	SPARE CIRCUIT	230V	0	0.0	2P	20 AT	0.0m	0.00%	SPARE PROVISION	-	SPARE
TOTAL	**CONNECTED DESIGN LOAD**	230V	**11450**	**35.9**	2P	**60 AT**	-	<= 3.0%	**2 - 14.0 mm2 THHN Cu**	**25 mm DIA Schedule 40 PVC**	**MAIN FEEDER (125% CONTINUOUS)**

FEEDER AMPACITY SIZING (PEC 2.15)

- Total Connected Load: 11450 VA (Net: 8250 VA)
- Full Load Current (I_net): 35.9 A @ 230V
- Continuous Factor: 125% Continuous Duty Factor
- Minimum Feeder Ampacity: 44.8 A Required
- Selected Conductor: 2 - 14.0 mm2 THHN Cu Cu (Ampacity: 65.0A @ 75C)

MAIN OVERCURRENT PROTECTION (PEC 2.40)

- Main Breaker: 60 AT / 100 AF, 2-Pole, 230V
- Interrupting Capacity: 10 kAIC Minimum RMS Symmetrical
- Trip Mechanism: Molded Case Thermal-Magnetic Inversed Time
- Enclosure Standard: NEMA-1 Surface Mounted Steel Cabinet
- Neutral Bus: 100A Insulated Solid Neutral Terminal Bar

EQUIPMENT GROUNDING SYSTEM (PEC 2.50)

- Grounding Conductor (GEC): 1 - 8.0 mm2 Bare Copper Wire
- Grounding Electrode: 16mm DIA x 2.4m Driven Copperweld Rod
- Ground Resistance Limit: Maximum 25 Ohms to Earth (PEC 2.50.3.17)
- Ground Bus: Solid Copper Grounding Terminal Bar Bonded to Box
- Inspection Well: Concrete Test Well with Removable Cover Plate

VOLTAGE DROP LIMITS & VERIFICATION (PEC 2.10.2.2(A) FPN 4)

- Maximum Branch Circuit Drop: <= 3.0% (Worst Branch: 2.12% - PASS)
- Maximum Main Feeder Drop: <= 2.0% (Calculated @ 15m: 0.75%)
- Combined Total System Drop: <= 5.0% (Total: 2.87% Feeder + Branch)
- Calculation Basis: VD = (2 x L x I x R) / 1000 at 75 deg C Copper
- Compliance Status: 100% PEC 2017 PART 1 COMPLIANT

2. SINGLE-LINE DIAGRAM (SLD) SCHEMATIC & SERVICE RISER (LP-1 (MAIN DISTRIBUTION PANEL))

VERTICAL RISER ELEVATION | SCALE: N.T.S.

PEC 2017 PART 1 COMPLIANT SERVICE

OVERHEAD SERVICE ENTRANCE (MERALCO)
230V AC, 1-PHASE, 2-WIRE, 60 Hz | PEC COMPLIANT

kWh — **REVENUE METER**
DIGITAL KILOWATT-HOUR REVENUE METER (MERALCO STANDARD)

60A — **MAIN CB: 60 AT / 100 AF, 2-Pole, 230V, 10 kAIC**
FEEDER: 2 - 14.0 mm2 THHN Cu IN 25 MM DIA SCHEDULE 40 PVC

PANEL: LP-1 (MAIN DISTRIBUTION PANEL) NEMA-1 SURFACE MOUNTED | 100A MAIN COPPER BUSBAR | 230V 1-PH 2-WIRE + GND

GROUND BUS

- 15A → **C1: Lighting Outlets (Living, Dining, Kitchen)**
  2 - 2.0 mm2 THHN Cu + 1 - 2.0 mm2 Ground in 20 mm DIA PVC
  VD: 0.82% [PASS] | 3D Run: 14.5m | **Load: 1200 VA (5.2A)**
- 20A → **C2: Convenience Outlets (Living & Master BR)**
  2 - 3.5 mm2 THHN Cu + 1 - 3.5 mm2 Ground in 20 mm DIA PVC
  VD: 1.45% [PASS] | 3D Run: 22.0m | **Load: 1800 VA (7.8A)**
- 30A → **C3: Air Conditioning Unit (Master BR - 1.5 HP)**
  2 - 5.5 mm2 THHN Cu + 1 - 5.5 mm2 Ground in 20 mm DIA PVC
  VD: 2.12% [PASS] | 3D Run: 38.5m | **Load: 2400 VA (10.4A)**
- 40A → **C4: Induction Cooktop / Range (Kitchen)**
  2 - 8.0 mm2 THHN Cu + 1 - 5.5 mm2 Ground in 25 mm DIA PVC
  VD: 1.68% [PASS] | 3D Run: 18.0m | **Load: 4500 VA (19.6A)**
- 20A → **C5: SPARE CIRCUIT**
  SPARE PROVISION in -

1 - 8.0 mm2 Bare Copper Wire
(R <= 25 Ohms, PEC 2.50.3.7)

3. STANDARD ELECTRICAL SYMBOLS & CAD LEGEND

[x]	Ceiling Lighting Outlet (100W Equivalent LED)	S3	Three-Way Wall Switch (16A, 250V, Multi-Point Control)
[o]	Recessed Downlight Ceiling Fixture (15W LED)	[CB]	Molded Case Circuit Breaker (Thermal-Magnetic Trip)
[=]	Duplex Convenience Receptacle (Grounding Type, 16A, 250V)	[LP-1]	Lighting and Power Distribution Panelboard Cabinet
[AC]	Air Conditioning Unit Special Receptacle (Dedicated 20A/30A)	[kWh]	Digital Kilowatt-Hour Revenue Meter (Utility Base)
[R]	Cooking Range / Induction Cooktop Outlet (50A, 250V)	[===]	Driven Copperweld Ground Rod with Ground Clamp
S1	Single-Pole Wall Switch (16A, 250V, Flush Mounted @ 1.37m)	==>	Homerun Branch Circuit Run to Panelboard LP-1

4. GENERAL ELECTRICAL SPECIFICATIONS & MANDATORY PEC 2017 NOTES

1. CODES & STANDARDS: All electrical works shall strictly comply with the Philippine Electrical Code (PEC 2017 Part 1), National Building Code of the Philippines, and regulations of the local electric utility and City Building Official.
2. SERVICE VOLTAGE: Service entrance voltage shall be 230 Volts AC, Single-Phase, 2-Wire, 60 Hz with solid copper equipment grounding conductor.
3. CONDUCTORS: All conductors shall be 99.9% electrolytic copper, 600V rated THHN/THWN-2 insulation (PEC Table 3.10.2.6(B)(16)). Minimum wire size shall be 2.0 mm2 (#14 AWG, max 15A OCPD) for lighting and 3.5 mm2 (#12 AWG, max 20A OCPD) for convenience outlets (PEC Sec. 2.40.1.4(D)).
4. RACEWAYS & CONDUITS: Raceways shall be Schedule 40 unplasticized polyvinyl chloride (uPVC) or EMT secured with clamps at not more than 900 mm (0.9 m) intervals for 15-25 mm PVC and within 900 mm (0.9 m) of each box (PEC Sec. 3.52.2.21 & Table 3.52.2.21).
5. MOUNTING HEIGHTS: Wall switches: 1.37m AFF; Convenience outlets: 0.30m AFF; Counter outlets: 1.05m AFF; Panelboard top: 1.80m AFF; kWh meter center: 1.52m AFF.
6. VOLTAGE DROP LIMITS: Feeder voltage drop shall not exceed 2.0%; branch circuit voltage drop shall not exceed 3.0%; combined system voltage drop shall not exceed 5.0% (PEC Sec. 2.10.2.2(A) FPN No. 4 & Sec. 2.15.1.2(A)(1) FPN No. 2).
7. GROUNDING SYSTEM: Provide 16mm dia x 2.4m copper-clad steel ground rod driven in inspection pit. Ground resistance shall not exceed 25 ohms (PEC Sec. 2.50.3.7). Equipment grounding conductor shall comply with PEC Table 2.50.6.13.
8. BOXES & FITTINGS: All junction and utility boxes shall have approved cover plates. Provide minimum 150mm (6") free conductor splice leads at each outlet, junction, and switch point (PEC Sec. 3.0.1.14).
9. TESTING & COMMISSIONING: Perform insulation resistance test (Megger 500V DC >= 1.0 Megohm minimum per PEC / IEEE 43) and earth ground resistance test (<= 25 Ohms) before final energization.
10. SUPERVISION: All electrical installations shall be executed under the direct supervision of a duly licensed Professional Electrical Engineer (PEE) or Registered Electrical Engineer (REE) per RA 7920.

5. SITE ELECTRICAL SERVICE CONNECTION & VICINITY PLAN

SITE ELECTRICAL SERVICE CONNECTION PLAN (SCALE: N.T.S.) [N] NORTH

P — MERALCO POLE SIN-4821

OVERHEAD SERVICE DROP (15.0m SPAN)
2 - 14.0 mm2 THHN Cu Cu + MESSENGER NEUTRAL

BUILDING FOOTPRINT
WEATHERHEAD & METER
PANELBOARD LP-1 INSIDE

PROJECT LOCATION DATA:

PROJECT: TWO-STOREY RESIDENTIAL BUILDING
ADDRESS: LOT 12 BLK 4, SAN LORENZO RUIZ, MAKATI CITY
CLIENT: ENGR. JUAN DELA CRUZ
ROAD R.O.W.: 10.00 METER ROAD RIGHT-OF-WAY
POWER DIST.: MERALCO 230V 1-PH SECONDARY RACK

BANAAG
ELECTRICAL ENGINEERING SUITE

PROFESSIONAL ELECTRICAL ENGINEER

PEE DRY SEAL HERE

Engr. Roberto Dela Rosa, PEE
PRC REG. NO.: 0000000 | PTR NO.: 0000000

PROJECT:
TWO-STOREY RESIDENTIAL BUILDING

LOCATION:
LOT 12 BLK 4, SAN LORENZO RUIZ, MAKATI CITY

OWNER / CLIENT:
ENGR. JUAN DELA CRUZ

DESIGNER:
BANAAG CAD Engine V14

CHECKED BY:
Engr. Roberto Dela Rosa, PEE

DATE / REVISION:
2026-10-23 (REV 0)

SHEET TITLE:
SINGLE LINE DIAGRAM & SCHEDULE OF LOADS

SHEET NO. **E-1**